AI-Powered Real-Time Solar Energy Generation Forecasting Using Weather-Aware Machine Learning Models and Explainable Artificial Intelligence

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Abstract

The increasing integration of renewable energy resources into modern power systems has created a growing demand for accurate forecasting methods capable of predicting power generation under dynamic environmental conditions. Among renewable energy sources, solar photovoltaic (PV) systems have emerged as one of the most promising technologies due to their sustainability, scalability, and declining installation costs. However, the intermittent nature of solar energy generation, which is strongly influenced by weather variability, presents significant challenges for grid operators and energy planners. Accurate forecasting of solar power generation is therefore essential for maintaining grid stability, reducing operational uncertainty, and improving energy management strategies.

This study proposes an Artificial Intelligence (AI)-powered solar energy forecasting framework that combines weather-aware machine learning models with explainable artificial intelligence techniques. Historical solar plant data consisting of power generation, irradiation, ambient temperature, and module temperature measurements were analyzed. Extensive preprocessing, feature engineering, and temporal feature extraction techniques were employed to improve predictive performance. Multiple forecasting models, including Linear Regression, Random Forest, LightGBM, XGBoost, Tuned XGBoost, and Long Short-Term Memory (LSTM) networks, were developed and compared.

Experimental results demonstrate that the Tuned XGBoost model achieved superior forecasting performance with an R^2 score of 0.9971, RMSE of 432.44 kW, and MAE of 215.38 kW, outperforming both conventional machine learning and deep learning approaches. SHAP-based explainability analysis identified irradiation, temperature difference, and temporal features as the most influential predictors. Furthermore, a real-time forecasting framework was developed using live weather data obtained through an API-based integration mechanism.

The proposed framework provides a practical, interpretable, and highly accurate solution for solar energy forecasting and demonstrates the potential of combining machine learning and explainable AI for next-generation renewable energy management systems.

I. Introduction

1.1 Background

The global energy sector is rapidly transitioning from fossil fuels to renewable energy sources due to rising electricity demand, environmental concerns, and the need to reduce greenhouse gas emissions. Among various renewable energy technologies, solar photovoltaic (PV) systems have become one of the most widely adopted solutions because they generate clean electricity and have experienced significant cost reductions in recent years. According to the International Energy Agency (IEA, 2024), solar energy is one of the fastest-growing renewable energy sources worldwide. [21]

Despite its advantages, solar power generation is highly dependent on weather conditions such as solar irradiance, temperature, humidity, and cloud cover. These environmental factors can change frequently, causing fluctuations in energy production. As a result, accurate solar energy forecasting has become essential for maintaining reliable and efficient power system operations.

1.2 Importance of Solar Energy Forecasting

As solar energy contributes a larger share of electricity generation, utility companies and grid operators face challenges in balancing energy supply and demand. Inaccurate forecasts may lead to power imbalances, increased operating costs, and reduced grid stability. Reliable forecasting enables better planning of electricity generation, energy storage management, and load distribution.

Accurate predictions also improve the economic performance of renewable energy systems by reducing uncertainty and optimizing resource allocation. Furthermore, forecasting supports the integration of renewable energy into smart grids and helps energy providers participate more effectively in electricity markets. Therefore, developing efficient forecasting models is crucial for sustainable energy management. [1], [10], [18]

1.3 Challenges in Solar Power Prediction

Solar power forecasting is a complex task because the relationship between weather variables and energy generation is highly nonlinear and dynamic. Factors such as changing cloud patterns, seasonal variations, temperature fluctuations, and incomplete sensor data can significantly affect forecasting accuracy. Traditional statistical methods, including linear regression and persistence models, often struggle to capture these complex relationships, resulting in higher prediction errors.

1.4 Machine Learning for Solar Forecasting

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have provided powerful solutions for renewable energy forecasting. Machine learning models can learn patterns from historical weather and power generation data, enabling them to make more accurate predictions. Algorithms such as Random Forest, XGBoost, and LightGBM have shown strong performance in capturing nonlinear relationships among environmental variables.

In addition, deep learning techniques such as Long Short-Term Memory (LSTM) networks can model time-dependent patterns in sequential data. These approaches have significantly improved forecasting accuracy compared to traditional methods and are increasingly being adopted in energy-related applications. [8], [9], [20]

1.5 Research Objectives and Contributions

This research aims to develop a weather-aware machine learning framework for accurate solar energy forecasting. The study evaluates multiple forecasting models, including Linear Regression, Random Forest, LightGBM, XGBoost, Tuned XGBoost, and LSTM. To improve transparency and trust, SHapley Additive exPlanations (SHAP) are used to interpret model predictions and identify the most influential weather factors.

The proposed framework also incorporates weather-based feature engineering and real-time forecasting using live weather data. By combining predictive accuracy with explainability, this study contributes to the development of reliable solar forecasting systems that support smart grids, renewable energy integration, and sustainable energy management.

II. Literature Review

2.1 Solar Energy Forecasting Research

The rapid growth of solar photovoltaic (PV) installations has increased the need for accurate forecasting systems. Since solar power generation depends heavily on weather conditions, fluctuations in solar output can create challenges for power system operators. Accurate forecasting helps maintain grid stability, optimize energy distribution, and reduce operational costs. As a result, solar energy forecasting has become a major area of research in renewable energy systems.

Early studies primarily focused on statistical forecasting techniques. However, researchers soon recognized that solar power generation is influenced by complex and nonlinear relationships among meteorological factors such as solar irradiance, temperature, humidity, cloud cover, and wind conditions. Antonanzas et al. (2016) provided a comprehensive review of solar forecasting methods and concluded that incorporating weather-related variables significantly improves prediction accuracy. Their findings highlighted the importance of integrating meteorological data into forecasting models.

Similarly, Mellit and Kalogirou (2008) explored the application of Artificial Intelligence (AI) in renewable energy systems and found that intelligent models were more effective than traditional statistical approaches when handling nonlinear data patterns. Their work laid the foundation for the increasing use of machine learning and deep learning techniques in solar forecasting research. [1], [9], [10], [18]

Further studies by Yang et al. (2018) emphasized the role of short-term solar forecasting in smart grid environments. They reported that weather-aware forecasting models can substantially reduce prediction errors and improve operational efficiency compared with traditional persistence-based forecasting methods.

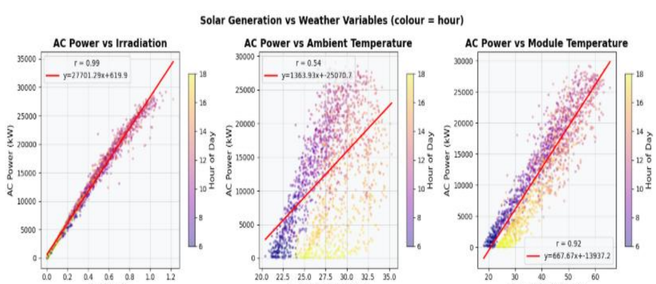


Figure 1. Power Generation vs Weather Variables — Scatter Plots

2.2 Machine Learning Approaches for Solar Forecasting

Machine learning has emerged as one of the most effective approaches for solar energy prediction because of its ability to learn complex relationships directly from historical data. Unlike traditional methods that rely on predefined assumptions, machine learning algorithms can automatically identify hidden patterns and interactions among variables.

One of the most widely used ensemble learning techniques is Random Forest, introduced by Breiman (2001). Random Forest combines multiple decision trees to improve predictive accuracy while reducing the risk of overfitting. Due to its robustness and ability to handle noisy datasets, it has been extensively applied in renewable energy forecasting.

Voyant et al. (2017) compared several forecasting techniques and found that ensemble-based models generally outperform traditional regression approaches. Their research demonstrated that combining predictions from multiple learners often leads to more stable and reliable forecasts.

Linear Regression remains a popular baseline model because of its simplicity and ease of interpretation. However, its performance is often limited when dealing with nonlinear relationships between weather conditions and solar energy generation. Wang et al. (2019) reported that linear models typically produce lower forecasting accuracy compared with advanced machine learning techniques.

The development of boosting algorithms marked a significant advancement in predictive modeling. Gradient Boosting Machines (GBM), proposed by Friedman (2001), introduced a sequential learning strategy that gradually improves model performance by correcting previous prediction errors. This concept later inspired highly successful algorithms such as XGBoost and LightGBM.

2.3 XGBoost and Advanced Ensemble Learning

Among modern machine learning techniques, XGBoost has gained widespread recognition for its exceptional predictive performance. Developed by Chen and Guestrin (2016), XGBoost incorporates regularization, efficient tree-building methods, and parallel processing capabilities, making it both accurate and computationally efficient.

Several studies have demonstrated the effectiveness of XGBoost in renewable energy forecasting. Wang et al. (2021) applied XGBoost to photovoltaic power prediction and observed substantial improvements over conventional machine learning models. The algorithm successfully captured complex interactions among weather variables, leading to more reliable forecasts.

Similarly, Zhang et al. (2022) used XGBoost for solar irradiance prediction and reported superior performance compared with Random Forest and Support Vector Regression models. Their findings highlighted the algorithm's ability to handle large datasets and nonlinear feature relationships effectively.

Another important ensemble learning algorithm is LightGBM, introduced by Ke et al. (2017). LightGBM improves computational efficiency through histogram-based learning and leaf-wise tree growth strategies. While it often trains faster than XGBoost, several studies suggest that XGBoost may provide slightly higher accuracy in photovoltaic forecasting tasks. Consequently, both algorithms are frequently evaluated in comparative forecasting studies.

2.4 Deep Learning in Renewable Energy Forecasting

Deep learning has become increasingly popular in energy forecasting due to its ability to model complex temporal patterns. Unlike traditional machine learning methods, deep learning architectures can automatically extract meaningful features from large datasets.

A major breakthrough in time-series forecasting came with the introduction of Long Short-Term Memory (LSTM) networks by Hochreiter and Schmidhuber (1997). LSTM networks were specifically designed to address the limitations of conventional recurrent neural networks by capturing long-term dependencies within sequential data. [5], [11], [12], [13]

Research by Shi et al. (2015) demonstrated that LSTM models perform well in renewable energy forecasting applications, particularly when temporal relationships play a significant role. Similarly, Qin et al. (2017) found that LSTM networks effectively capture seasonal and hourly solar generation patterns.

Despite their advantages, deep learning models often require large amounts of training data and significant computational resources. Recent studies indicate that well-optimized ensemble learning methods such as XGBoost and LightGBM can achieve comparable or even superior performance on medium-sized datasets while requiring less computational effort.

2.5 Explainable Artificial Intelligence in Energy Forecasting

Although advanced machine learning models provide highly accurate predictions, they are often criticized for their lack of transparency. This challenge is particularly important in energy systems, where decision-makers need to understand the reasoning behind model outputs before deploying them in real-world environments.

Explainable Artificial Intelligence (XAI) addresses this issue by making model predictions more interpretable. One of the most widely adopted explainability techniques is SHapley Additive exPlanations (SHAP), introduced by Lundberg and Lee (2017). SHAP assigns contribution scores to input features, allowing researchers to understand how each variable influences a prediction.

According to Molnar (2022), explainability techniques improve stakeholder trust and support model validation. In solar forecasting applications, SHAP has been used to identify the influence of key variables such as solar irradiance, temperature, humidity, and time-based features. These insights help researchers verify model behavior and ensure that predictions align with real-world physical principles.

2.6 Research Gap Analysis

Although significant progress has been made in solar energy forecasting, several research gaps remain. Many existing studies focus primarily on improving prediction accuracy while giving limited attention to model explainability. Similarly, some forecasting frameworks rely mainly on historical power generation data without incorporating comprehensive weather-aware feature engineering.

Another limitation is the lack of integrated systems that combine advanced machine learning models, explainable AI techniques, and real-time weather information. While deep learning approaches have received considerable attention, recent evidence suggests that optimized ensemble learning algorithms may offer comparable or better performance with lower computational costs.

Therefore, there is a need for a comprehensive framework that combines weather-based feature engineering, advanced machine learning algorithms, deep learning comparison, SHAP-based explainability, and real-time forecasting capabilities. This study aims to address these gaps by developing an accurate, interpretable, and practical solar energy forecasting system suitable for modern renewable energy management applications.

III. Methodology

3.1 Research Framework

This study proposes a weather-aware Artificial Intelligence (AI) framework for forecasting solar photovoltaic (PV) power generation. The methodology consists of six main stages: data collection, data preprocessing, feature engineering, model development, model evaluation, and explainability analysis. Historical solar power generation data and weather-related variables were used to train predictive models capable of estimating future energy output.

The workflow begins with collecting and preparing the dataset, followed by generating meaningful temporal and environmental features. Multiple machine learning and deep learning models were then trained and compared using standard evaluation metrics. Finally, SHAP-based explainability analysis and a real-time forecasting module were implemented to improve model transparency and practical usability.

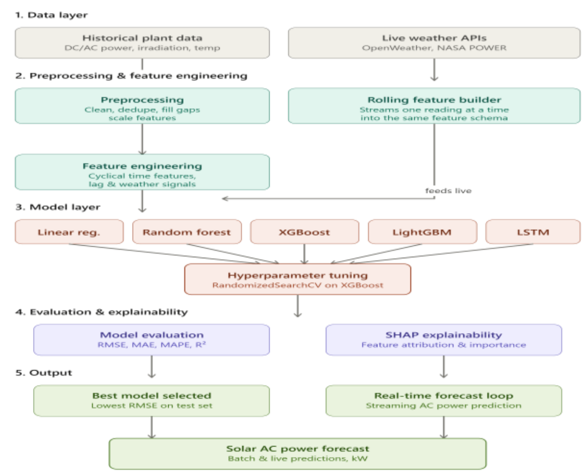


Figure 2. Proposed AI-Powered Solar Forecasting Framework

3.2 Dataset Description

The dataset consists of historical records from a solar photovoltaic power plant and includes both energy generation measurements and environmental factors that influence solar panel performance. The target variable used for prediction is AC_POWER, which represents the actual electrical power generated by the system.

Feature	Description
DATE_TIME	Timestamp of observation
AC_POWER	Actual AC power output (Target Variable)
DC_POWER	Generated DC power
IRRADIATION	Solar irradiance received by panels
AMBIENT_TEMPERATURE	Surrounding air temperature
MODULE_TEMPERATURE	Solar panel surface temperature

Table 1. Dataset Features

3.3 Data Preprocessing

Data preprocessing was performed to ensure data quality and improve model performance. Missing values were handled using appropriate statistical methods, while abnormal observations were identified and analyzed through visualization techniques. The timestamp column was converted into datetime format to enable extraction of time- related information.

For deep learning implementation, numerical variableswere normalized using Min-Max Scaling to ensure efficient model training and faster convergence.

3.4 Feature Engineering

Feature engineering was used to create meaningful variables that improve forecasting accuracy. Time-based features such as hour, day, month, and weekday were extracted from the timestamp to capture daily and seasonal generation patterns.

Since time variables are cyclical in nature, sine and cosine transformations were applied:

$$Hour_{sin} = sin\left(\frac{2\pi \times Hour}{24}\right)$$
$$Hour_{cos} = cos\left(\frac{2\pi \times Hour}{24}\right)$$

A temperature difference feature was also generated to represent thermal effects on panel efficiency:

$$TempDiff = ModuleTemperature - AmbientTemperature$$

Historical power generation information was incorporated using lag features:

$$Lag_1 = Power_{t-1}$$
$$Lag_{24} = Power_{t-24}$$

Additionally, rolling averages were calculated to capture short-term trends:

$$RollingMean = \frac{1}{n}\sum_{i=1}^{n} x_i$$

These engineered features helped the models learn both temporal and environmental influences on solar energy production.

3.5 Model Development

To identify the most effective forecasting approach, six predictive models were implemented and compared:

- Linear Regression – used as a baseline model due to its simplicity and interpretability.
- Random Forest – an ensemble learning method capable of capturing nonlinear relationships.
- LightGBM – a gradient boosting algorithm known for fast training and efficient performance.
- XGBoost – an advanced boosting algorithm with regularization mechanisms to reduce overfitting.
- Tuned XGBoost – optimized using Grid Search Cross Validation to achieve improved predictive accuracy.
- Long Short-Term Memory (LSTM) – a deep learning model designed to capture sequential and time-dependent patterns in solar generation data.

The tuned XGBoost model utilized optimized parameters including 300 estimators, maximum depth of 4, learning rate of 0.05, subsample ratio of 0.8, and column sampling ratio of 0.9.

3.6 Model Evaluation

The dataset was divided into training and testing sets using an 80:20 ratio. Model performance was evaluated using four standard regression metrics:

Mean Absolute Error (MAE)

$$MAE = \frac{1}{n}\sum_{i=1}^{n} | y_i - \hat{y}_i |$$

Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{100}{n}\sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

These metrics were used to assess prediction accuracy and compare model performance.

3.7 Explainability and Real-Time Forecasting

To improve model transparency, SHapley Additive exPlanations (SHAP) were applied to the best-performing model. SHAP identifies the contribution of each feature to individual predictions, helping explain how environmental and temporal variables influence solar power generation.

A real-time forecasting module was also developed using weather data obtained through the OpenWeather API. Incoming weather information was processed using the same feature engineering pipeline, and the optimized XGBoost model generated live solar power forecasts. This demonstrates the practical applicability of the proposed framework for smart grid operations and renewable energy management systems.

IV. System Architecture and Implementation

4.1 System Architecture

The proposed AI-powered solar energy forecasting framework was developed as a multi-stage system that converts historical photovoltaic and weather data into accurate power generation forecasts. The architecture integrates data preprocessing, feature engineering, machine learning, explainable AI, and real-time forecasting within a single workflow.

The process begins with historical solar plant and meteorological data collection. After cleaning and preprocessing, relevant temporal and environmental features are generated to improve predictive performance. The processed dataset is divided into training and testing sets, and multiple forecasting models—including Linear Regression, Random Forest, LightGBM, XGBoost, Tuned XGBoost, and LSTM—are trained and evaluated. The best-performing model is then deployed for real-time forecasting using live weather data obtained through the OpenWeather API.

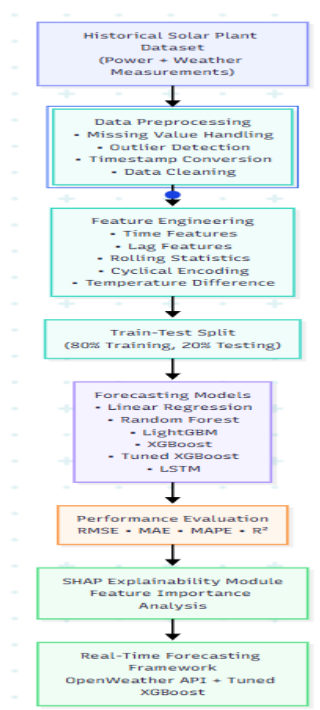


Figure 3. Proposed AI-Powered Solar Forecasting Architecture

4.2 Implementation Environment

The framework was implemented in Python using Jupyter Notebook. Several open-source libraries were utilized, including Pandas and NumPy for data processing, Matplotlib and Seaborn for visualization, Scikit-Learn for machine learning, XGBoost and LightGBM for ensemble learning, TensorFlow/Keras for deep learning, and SHAP for model interpretability. The use of open-source tools ensures reproducibility and scalability.

4.3 Model Development and Explainability

Six predictive models were developed and compared using standard evaluation metrics. Hyperparameter tuning was performed using Grid Search Cross Validation, resulting in an optimized XGBoost model with 300 trees, maximum depth of 4, learning rate of 0.05, subsample ratio of 0.8, and column sampling ratio of 0.9. This model achieved the highest forecasting accuracy.

To improve transparency, SHAP (SHapley Additive exPlanations) was integrated into the framework. The analysis revealed that solar irradiation, temperature-related variables, and temporal features had the greatest influence on power generation predictions.

Figure 4. Dataset Correlation Heatmap

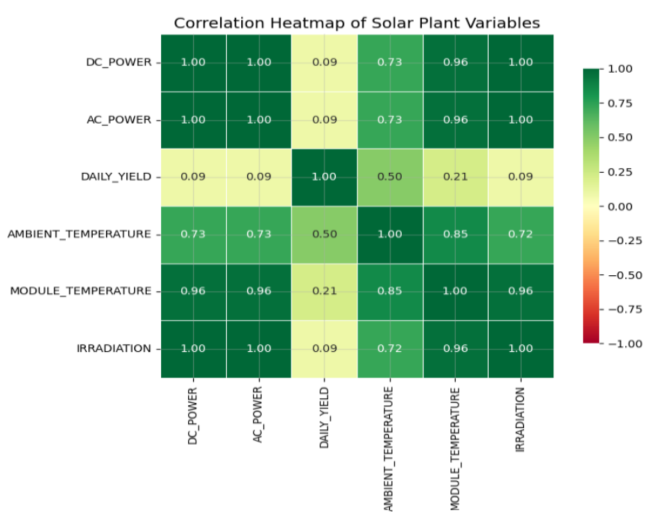
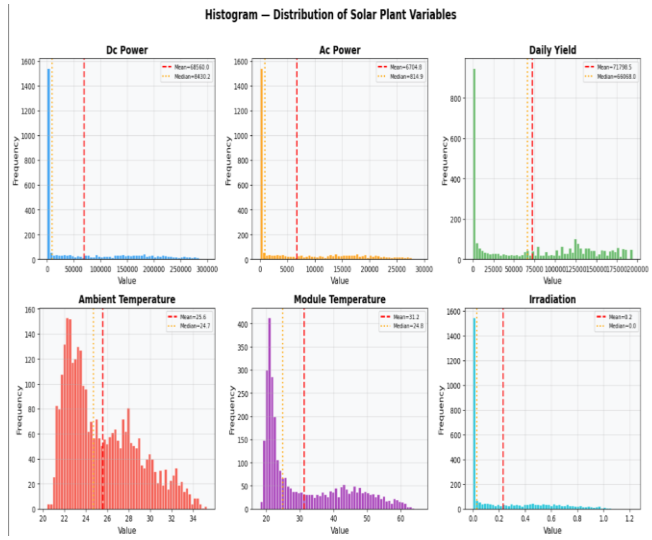


Figure 5. Feature Distribution Analysis



4.4 Real-Time Forecasting

A real-time forecasting module was developed using the OpenWeather API. Live weather observations are processed through the same feature engineering pipeline used during training, and the optimized XGBoost model generates continuous solar power forecasts. This enables practical deployment in smart grid and renewable energy management applications.

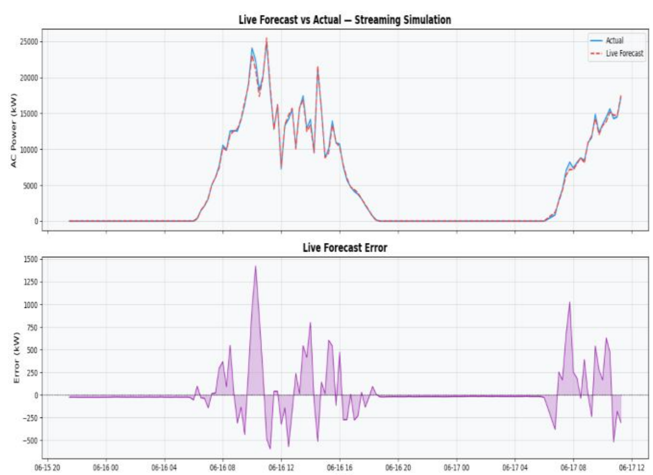


Figure 6. live forecast

V. Results and Discussion

5.1 Experimental Results

The proposed solar energy forecasting framework was evaluated using historical photovoltaic generation data and corresponding weather measurements. The dataset was divided into training and testing sets using an 80:20 ratio, and all models were trained on the same feature set to ensure a fair comparison. Model performance was assessed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and the Coefficient of Determination (R^2).

Among the six evaluated models, the ensemble learning approaches consistently outperformed the baseline Linear Regression and LSTM models. The Tuned XGBoost model achieved the best overall performance with an RMSE of 432.44 kW, MAE of 215.38 kW, and an R^2 score of 0.9971. These results indicate that the model successfully explained approximately 99.71% of the variation in photovoltaic power generation.

Random Forest and standard XGBoost also demonstrated strong forecasting capabilities, achieving R^2 values above 0.996. In contrast, Linear Regression produced higher prediction errors due to its inability to capture the complex nonlinear relationships between weather conditions and solar power output.

5.2 Impact of Hyperparameter Optimization

Hyperparameter tuning significantly improved XGBoost performance. Using Grid Search Cross Validation, the optimal configuration was identified with 300 trees, maximum depth of 4, learning rate of 0.05, subsample ratio of 0.8, and column sampling ratio of 0.9. The optimized model reduced forecasting errors and provided better generalization on unseen data, making it the most reliable forecasting approach in this study.

```
=====================================================================
  FINAL RESULTS SUMMARY — AI-Powered Solar Generation Forecasting
=====================================================================
               Model      RMSE       MAE     MAPE      R2
1    XGBoost (Tuned) 432.4401 215.3814   8.1769 0.9971
2            XGBoost 455.6608 223.2964   7.0286 0.9967
3      Random Forest 457.9502 219.4752   4.7449 0.9967
4           LightGBM 503.9223 249.0661   6.7190 0.9960
5  Linear Regression 581.9188 322.1600 63.4548 0.9947
6               LSTM 771.8870 458.2803 60.9428 0.9906
=====================================================================

 Best Model    : XGBoost (Tuned)
   RMSE = 432.4401 kW  |  R² = 0.9971
```

Figure 7. Model Performance Comparison

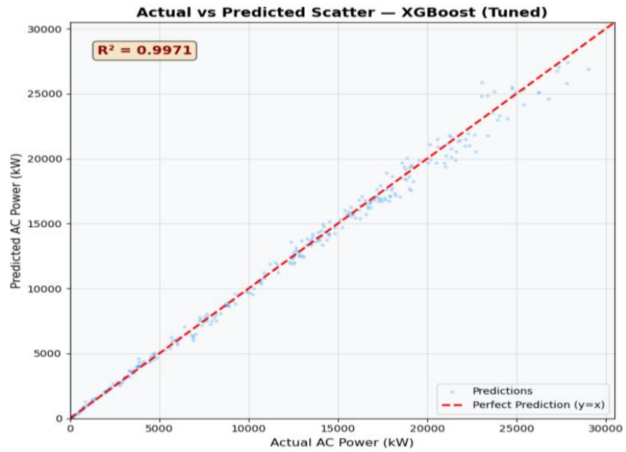


Figure 8. Actual vs Predicted (Best Model)

5.3 Explainability and Real-Time Forecasting

To improve model transparency, SHAP analysis was applied to the Tuned XGBoost model. The results revealed that solar irradiation was the most influential feature affecting power generation, followed by temperature difference, module temperature, ambient temperature, and temporal variables. These findings align with established photovoltaic principles, confirming the physical relevance of the model.

The optimized model was further integrated with live weather data obtained through the OpenWeather API for real-time forecasting. The framework achieved satisfactory performance under dynamic environmental conditions, demonstrating its practical applicability for smart grids and renewable energy management systems.

Overall, the results confirm that combining weather-aware feature engineering, optimized ensemble learning, explainable AI, and real-time data integration can provide highly accurate and reliable solar power forecasts.

```
=====================================================
  LIVE FORECAST — powered by OpenWeather
=====================================================
  Timestamp          : 2026-06-22 18:33:58.060283
  Irradiation (0-1)  : 0.169
  Ambient Temp (°C)  : 40.40
  Module Temp (°C)   : 42.43
  Cloud Cover (%)    : 4
  Conditions         : clear sky
-----------------------------------------------------
  Forecasted AC Power : 5336.45 kW
=====================================================
```

Figure 9. Live Forecasting

VI. Conclusion and Future Work

6.1 Achievements

This research successfully developed an AI-powered solar energy forecasting framework that combines weather-aware machine learning, feature engineering, explainable artificial intelligence, and real-time forecasting capabilities. Historical photovoltaic and meteorological data were used to train and evaluate multiple forecasting models, including Linear Regression, Random Forest, LightGBM, XGBoost, Tuned XGBoost, and LSTM.

The experimental results demonstrated that ensemble learning methods outperformed both traditional statistical approaches and deep learning models. Among all evaluated techniques, the Tuned XGBoost model achieved the best performance with an R^2 score of 0.9971, indicating excellent forecasting accuracy. The integration of SHAP explainability further improved model transparency by identifying the most influential factors affecting solar power generation. Additionally, the successful implementation of a real-time forecasting module using live weather data highlights the practical applicability of the proposed framework in renewable energy management systems.

6.2 Limitations

Despite achieving strong forecasting performance, several limitations were identified. The study relied on a single photovoltaic dataset, which may limit the generalizability of the results across different geographical regions and climatic conditions. Furthermore, only a limited number of weather variables were considered. Factors such as cloud cover, wind speed, humidity, and atmospheric pressure could potentially improve forecasting accuracy. The real-time forecasting module was also dependent on external weather APIs, making prediction quality reliant on the availability and accuracy of weather data.

6.3 Answers to Research Questions

The findings of this study provide clear answers to the research questions. First, machine learning models can accurately forecast solar power generation when combined with weather-aware features and historical generation data. Second, optimized ensemble learning algorithms, particularly Tuned XGBoost, outperform traditional regression and deep learning approaches for the selected dataset. Third, explainable AI techniques such as SHAP can successfully identify the key factors influencing model predictions, improving transparency and trust. Finally, integrating live weather data enables effective real-time forecasting suitable for practical deployment.

6.4 Future Work

Future research can enhance the framework by incorporating additional meteorological variables, larger datasets, and multiple solar plant locations. Advanced deep learning architectures such as GRU, Transformer, and Temporal Fusion Transformer models may also be explored. Furthermore, integrating satellite imagery and weather radar data could improve short-term forecasting accuracy. From an operational perspective, deploying the system on cloud and IoT-based platforms would enable continuous monitoring and automated decision-making within smart grid environments. These improvements could further strengthen the role of AI-driven forecasting in supporting sustainable and intelligent energy systems.

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